15/10/26

Lab 4: Ant Codony optimization:

Algorithm / Meaning:

- When ants have codonies or made homes they need to find optimal paths between codonies, so they while travelling they leave pheramone on the path travelled
- for the shortest path where multiple ants move through the pheramone will have higher scent strength and can be most travelled
- ants find shortest path in this having most pheramone

Pseudo Code:

```
Def ACO-TSP (D, n, m, MaxIterations):
    pheramone = initialize-pheramone(n)
    Cost distance = float (inf)
    best tour = initialize empty
    for iteration in range (MaxIterations):
        for k in range (m):
            unvisited (empty)
            Start = random-city (n)
            Tour [k] = [Start]
            unvisited = rem-cities (n, Start)
            while unvisited:
                Prob = calc-prob (next, st, n)
                next-city = select-city (prob)
                Tour [k]. append (next-city)
                unvisited. remove (next-city)
                Best-distance = calc-distance (next, D)
    pheramone * = (-1) (distance)
```

```
        if Tour Distance[k] < Best - Distance:
            Best - Distance = Tourdist[k]
            Best - tour = Tour[k]

    for k in range(m):
        for (i,j) in range consec-edges(Tour[k]):
            Pheramone[i][j] += Q/Tour-dist

    return Best tour in Best distance:
```

Proceed

Routing Pblms.

Paper: ACO based routing in wireless sensor net

by: Xuxun Liu (2017)

Summary: This paper presents a comprehensive survey of ACO based routing protocols for wireless sensors It classifies these protocols according to operating bh -our, primary objectives and topology. It discuss repre methods compares characteristics and performance com -ation & QOS including delay and reliability

- it provides researcher & network designer with way to understand compare various scarons & domains and available solutions

Objective :

- The objective of the paper is to systematically review, classify, qualitatively compare existing ACO based routing for wireless networks, highlight advantages, limitations and performance considerations
- identify research problems & give guidance for selecting suitable protocols for applications

Code :

```
edges = [ (0,1,2), (0,2,3), (1,3,2),
          (2,3,1), (1,4,4), (3,4,2),
          (3,5,3), (4,6,2), (5,6,2)]
```

Gr.add-edge (u,v, distance = distance)

-edge (u,v)

Takes two node ids as input and returns a key

route-cost (path)

takes route as input and calculates total path

Construct-path ()

use pheromone values and link distances to select neighbor

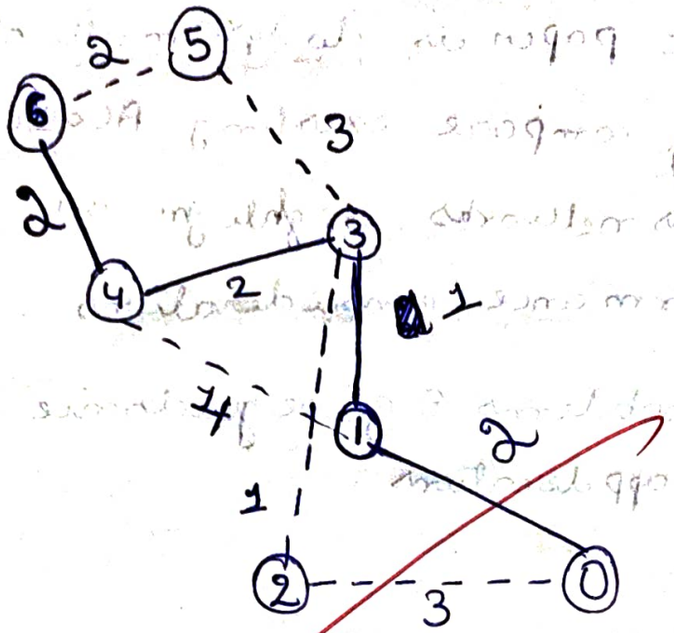
ACO-routing ()

runs as iterations, evaluate & store best

evaporate-pheromone ()

remove all olders and deposit selected route

o/p



Best route : [0, 1, 3, 4, 6]

Total : 8

o/p verified
COV
15/10/26.